

Minimization problems

There are two alternatives:

1. Maximize the symmetric of the objective function:

$$\min z = \mathbf{c}'\mathbf{x} \quad \Leftrightarrow \quad \max z' = -\mathbf{c}'\mathbf{x}$$

2. Change the optimality test and the criteria used to choose the variable entering the basis

In this curricular unit we use the **first alternative**

Example

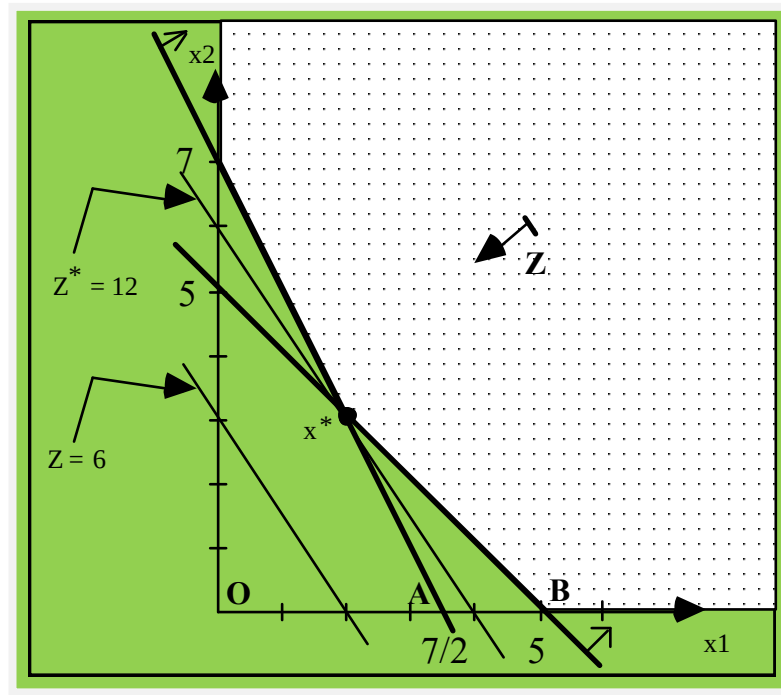
$$\min z = 3x_1 + 2x_2$$

subject to

$$x_1 + x_2 \geq 5$$

$$2x_1 + x_2 \geq 7$$

$$x_1 \geq 0, x_2 \geq 0$$



Firstly, the objective function is transformed into a maximization:

$$\min z = 3 x_1 + 2 x_2 \Leftrightarrow \max z' = -3 x_1 - 2 x_2$$

Then, the Simplex method is applied (using the “Two Phases” technique):

1st Phase

$$\max z_{1st\ phase} = - x_5 - x_6$$

$\uparrow \quad \uparrow$
artificial

subject to

$$x_1 + x_2 - x_3 + x_5 = 5$$

$$2 x_1 + x_2 - x_4 + x_6 = 7$$

$$x_i \geq 0, i = 1, 2, \dots, 6$$

Initial table (of 1st phase)

C_i	0	0	0	0	-1	-1	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	b
$x_5 \ -1$	1	1	-1	0	1	0	5
$\leftarrow x_6 \ -1$	<u>2</u> *	1	0	-1	0	1	7
$Z_j - c_j$	-3	-2	1	1	0	0	-12

↑

POINT O → not feasible

C_i	0	0	0	0	-1	-1	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	b
$\leftarrow x_5 \ -1$	0	1/2	-1	<u>1/2</u> *	1	-1/2	3/2
$x_1 \ 0$	1	1/2	0	-1/2	0	1/2	7/2
$Z_j - c_j$	0	-1/2	1	-1/2	0	3/2	-3/2

↑

POINT A → not feasible

Ci		0	0	0	0	-1	-1	b
x_B	$c'_B x_i$	x_1	x_2	x_3	x_4	x_5	x_6	
x_4	0	0	1	-2	1	2	-1	3
x_1	0	1	1	-1	0	1	0	5
$Z_j - c_j$		0	0	0	0	1	1	0

Optimal table of the 1st phase because there are no negative values in the row $z_j - c_j$

Optimal solution of the 1st phase:

BVs

$$x_1 = 5$$

$$x_4 = 3$$

NBVs

$$x_2 = 0$$

$$x_3 = 0$$

$$x_5 = 0$$

$$x_6 = 0$$

x_5 and x_6 are NBVs → feasible basic solution for the initial problem → POINT B

2nd Phase

$$\max z_{2\text{nd phase}} = -3x_1 - 2x_2$$

Initial table (of 2nd phase)

C_i		-3	-2	0	0	b
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	
← x_4	0	0	<u>1</u> *	-2	1	3
x_1	-3	1	1	-1	0	5
$Z_j - C_j$		0	-1	3	0	-15
		↑				

C_i		-3	-2	0	0	b
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	
x_2	-2	0	1	-2	1	3
x_1	-3	1	0	1	-1	2
$Z_j - C_j$		0	0	1	1	-12

Optimal table of the 2nd phase because there are no negative values in the row $z_j - c_j$

Optimal solution of the problem:

BVs	NBVs
$x_1 = 2$	$x_3 = 0$
$x_2 = 3$	$x_4 = 0$

The optimal solution corresponds to POINT $\mathbf{x}^*$.

Since we were maximizing z' , the optimal value that we retrieve from the Simplex table is $\mathbf{z}'^*$.
The optimal value of z will be obtained in the following way:

$$z'^* = -12 \Rightarrow z^* = 12$$

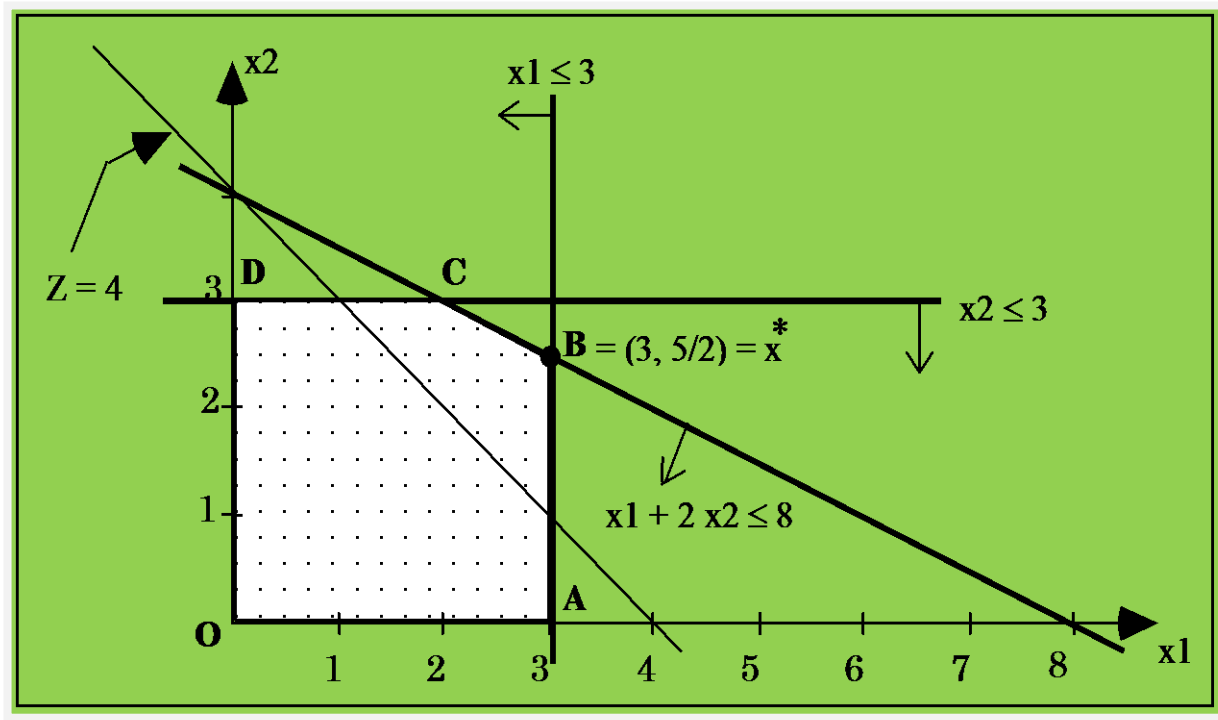
Particular cases of Simplex method

- The same value of $z_j - c_j$ for different j
 - ***any one can be used***
(the only difference is in the followed “way” to reach the optimal)
- There exists a negative $z_j - c_j$ value but in the pivot column there are only non-positive values
 - ***unbounded solution***
- An artificial variable appears in the optimal solution
 - ***there is no solution***
(there is not a feasible region)
- A tie in the choice of the pivot element
 - ***any one can be used but a degenerate solution will be obtained***
(basic variable equal to zero)
- $Z_j - c_j$ is equal to zero but x_j is a NBV
 - ***alternative optimal solutions***

The same value of $Z_j - c_j$ for different j

Example

$$\begin{aligned} \max \quad & z = x_1 + x_2 \\ \text{subject to} \quad & x_1 \leq 3 \\ & x_2 \leq 3 \\ & x_1 + 2x_2 \leq 8 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$



Solving by the Simplex method:

$$\max z = x_1 + x_2$$

subject to

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 3$$

$$x_1 + 2x_2 + x_5 = 8$$

$$x_i \geq 0; i = 1, 2, \dots, 5$$

Initial table

C_i		1	1	0	0	0	b	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5		
x_3	0	1	0	1	0	0	3	$x_1 = 0$
x_4	0	0	1	0	1	0	3	$x_2 = 0$
x_5	0	1	2	0	0	1	8	$x_3 = 3$
$Z_j - C_j$		-1	-1	0	0	0	0	$x_4 = 3$
		↑	↑					$x_5 = 8$

Any of x_1 or x_2 can enter the basis

$$Z = 0$$

POINT O

- If x_1 is selected then x_3 will leave the basis and the next feasible basic solution corresponds to point **A**
- If x_2 is selected then x_4 will leave the basis and the next feasible basic solution corresponds to point **D**

Choosing the first alternative, the obtained table is the following:

C_i		1	1	0	0	0		
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	x_5	b	
x_1	1	1	0	1	0	0	3	$x_1 = 3$
x_4	0	0	1	0	1	0	3	$x_2 = 0$
$\leftarrow x_5$	0	0	<u>2</u> *	-1	0	1	5	$x_3 = 0$
$Z_j - c_j$		0	-1	1	0	0	3	$x_4 = 3$
								$x_5 = 5$
								$Z = 3$

POINT A

C_i		1	1	0	0	0		
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	x_5	b	
x_1	1	1	0	1	0	0	3	$x_1 = 3$
x_4	0	0	0	1/2	1	-1/2	1/2	$x_2 = 5/2$
x_2	1	0	1	-1/2	0	1/2	5/2	$x_3 = 0$
$Z_j - c_j$		0	0	1/2	0	1/2	11/2	$x_4 = 1/2$
								$x_5 = 0$
								$Z = 11/2$

Optimal solution $\rightarrow$ POINT B

Choosing the second alternative, the obtained table is the following:

C_i		1	1	0	0	0		
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	x_5	b	
x_3	0	1	0	1	0	0	3	$x_1 = 0$
x_2	1	0	1	0	1	0	3	$x_2 = 3$
$\leftarrow x_5$	0	<u>1</u> *	0	0	-2	1	2	$x_3 = 3$
$Z_j - c_j$		-1	0	0	1	0	3	$x_4 = 0$
								$x_5 = 2$
								$Z = 3$

POINT D

C_i	1	1	0	<u>0</u>	0		
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	b	
$\leftarrow x_3 \ 0$	0	0	1	<u>2</u> *	-1	1	$x_1 = 2$
$x_2 \ 1$	0	1	0	1	0	3	$x_2 = 3$
$x_1 \ 1$	1	0	0	-2	1	2	$x_3 = 1$
$Z_j - c_j$	0	0	0	-1	1	5	$x_4 = 0$
				$\uparrow$			$x_5 = 0$
							$Z = 5$

POINT C

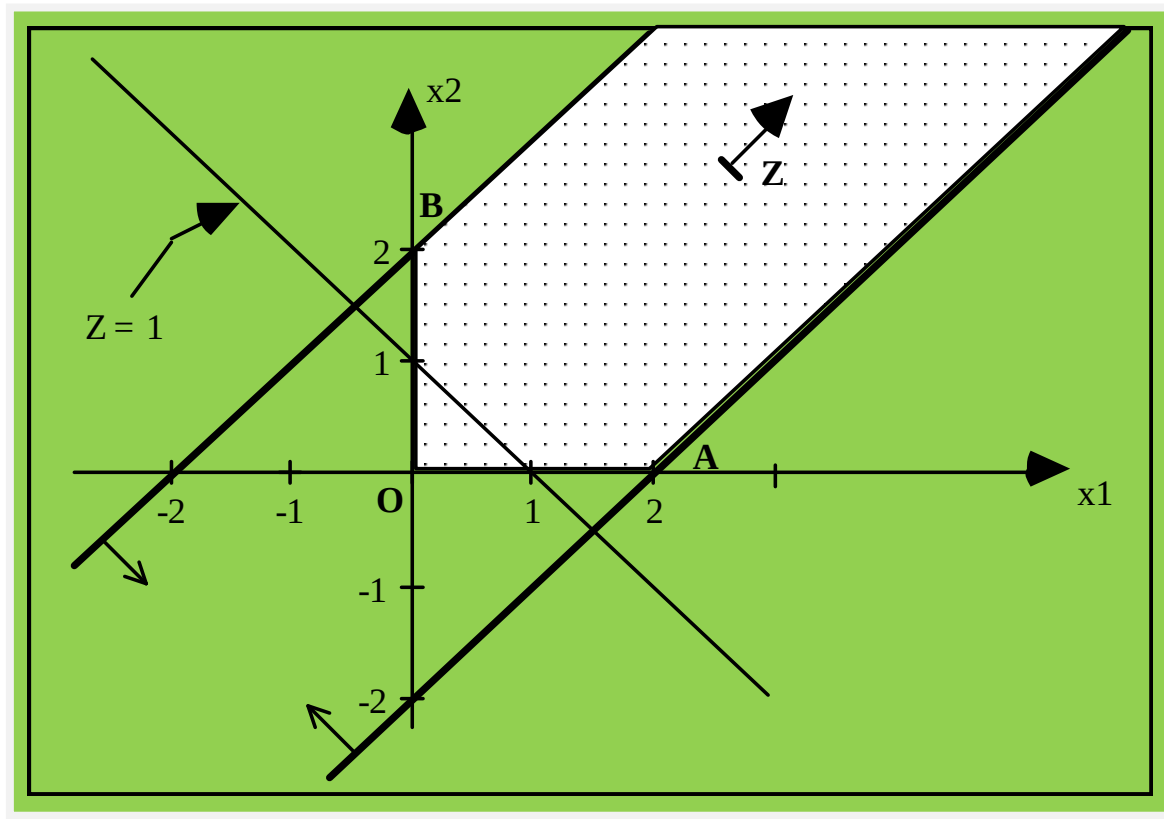
C_i	1	1	0	0	0		
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	b	
$x_4 \ 0$	0	0	1/2	1	-1/2	1/2	$x_1 = 3$
$x_2 \ 1$	0	1	-1/2	0	1/2	5/2	$x_2 = 5/2$
$x_1 \ 1$	1	0	1	0	0	3	$x_3 = 0$
$Z_j - c_j$	0	0	1/2	0	1/2	11/2	$x_4 = 1/2$
							$x_5 = 0$
							$Z = 11/2$

Optimal solution $\rightarrow$ POINT B

Unbounded solution

Example

$$\begin{aligned} \max \quad & z = x_1 + x_2 \\ \text{subject to} \quad & x_1 - x_2 \leq 2 \\ & -x_1 + x_2 \leq 2 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$



Solving by the Simplex method:

$$\max z = x_1 + x_2$$

subject to

$$x_1 - x_2 + x_3 = 2$$

$$-x_1 + x_2 + x_4 = 2$$

$$x_i \geq 0; i = 1, 2, 3, 4$$

Initial table

C_i		1	1	0	0	b	
x_B	C'_B	x_1	x_2	x_3	x_4		
$\leftarrow x_3$	0	<u>1</u> *	-1	1	0	2	$x_1 = 0$
x_4	0	-1	1	0	1	2	$x_2 = 0$
$Z_j - C_j$		-1	-1	0	0	0	$x_3 = 2$
		$\uparrow$					$x_4 = 2$
							$Z = 0$

POINT O

C_i	1	1	0	0		
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	b	
(?) $x_1 \ 1$	1	-1	1	0	2	$x_1 = 2$
(?) $x_4 \ 0$	0	0	1	1	4	$x_2 = 0$
$Z_j - C_j$	0	-2	1	0	2	$x_3 = 0$
		↑				$x_4 = 4$
						$Z = 2$

POINT A

Optimal solution is unbounded

Inexistent solution

Example

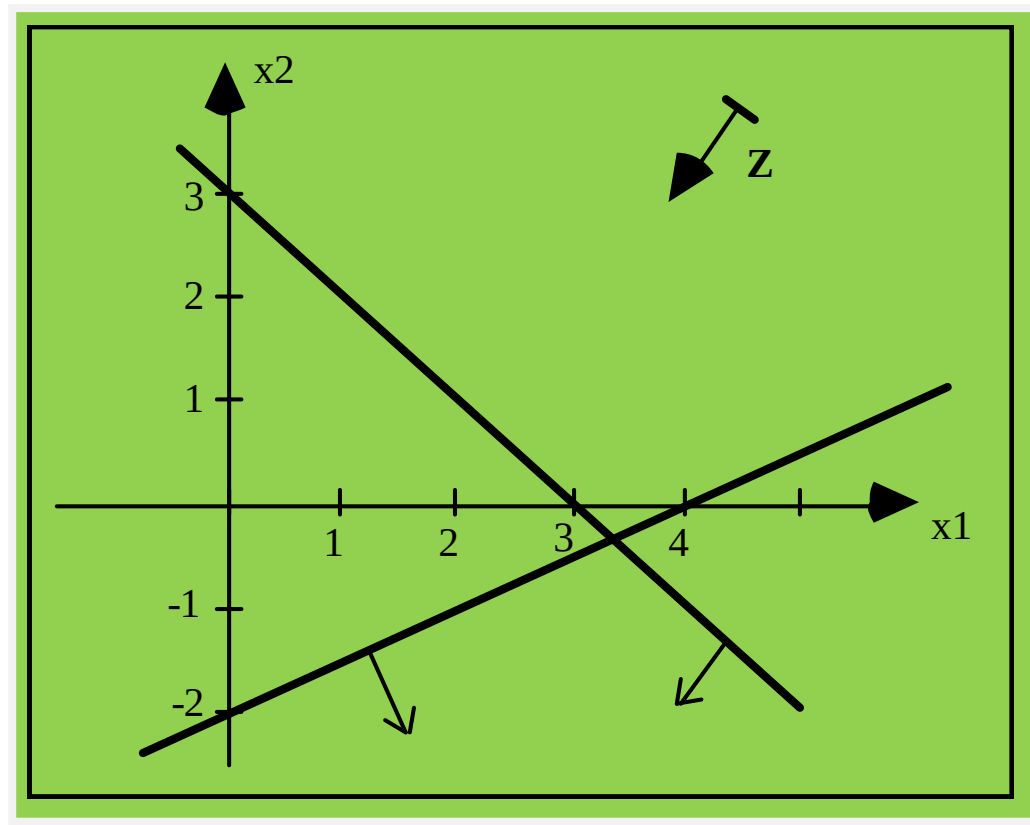
$$\min z = 3x_1 + 4x_2$$

subject to

$$x_1 - 2x_2 \geq 4$$

$$x_1 + x_2 \leq 3$$

$$x_1 \geq 0, x_2 \geq 0$$



Solving by the Simplex method (with the “Two Phases “ technique):

1st Phase

$$\max z_{1^{\text{st phase}}} = -x_4$$

subject to

$$x_1 - 2x_2 - x_3 + x_4 = 4$$

$$x_1 + x_2 + x_5 = 3$$

$$x_i \geq 0; i = 1, 2, \dots, 5$$

Initial table

C_i		0	0	0	-1	0	b
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	
x_4	-1	1	-2	-1	1	0	4
← x_5	0	<u>1</u> *	1	0	0	1	3
$Z_j - C_j$		-1	2	1	0	0	-4

↑

ci		0	0	0	-1	0	b
$x_B \quad c'_B \quad x_i$		x_1	x_2	x_3	x_4	x_5	
x_4	-1	0	-3	-1	1	-1	1
x_1	0	1	1	0	0	1	3
$Z_j - c_j$		0	3	1	0	1	-1

Optimal table of the 1st phase with:

$$Z_{1st \text{ phase}} = -1 < 0 \quad \rightarrow \quad x_4 \text{ (artificial variable) is in the basis}$$

The obtained solution is not feasible



Inexistent solution (impossible problem)

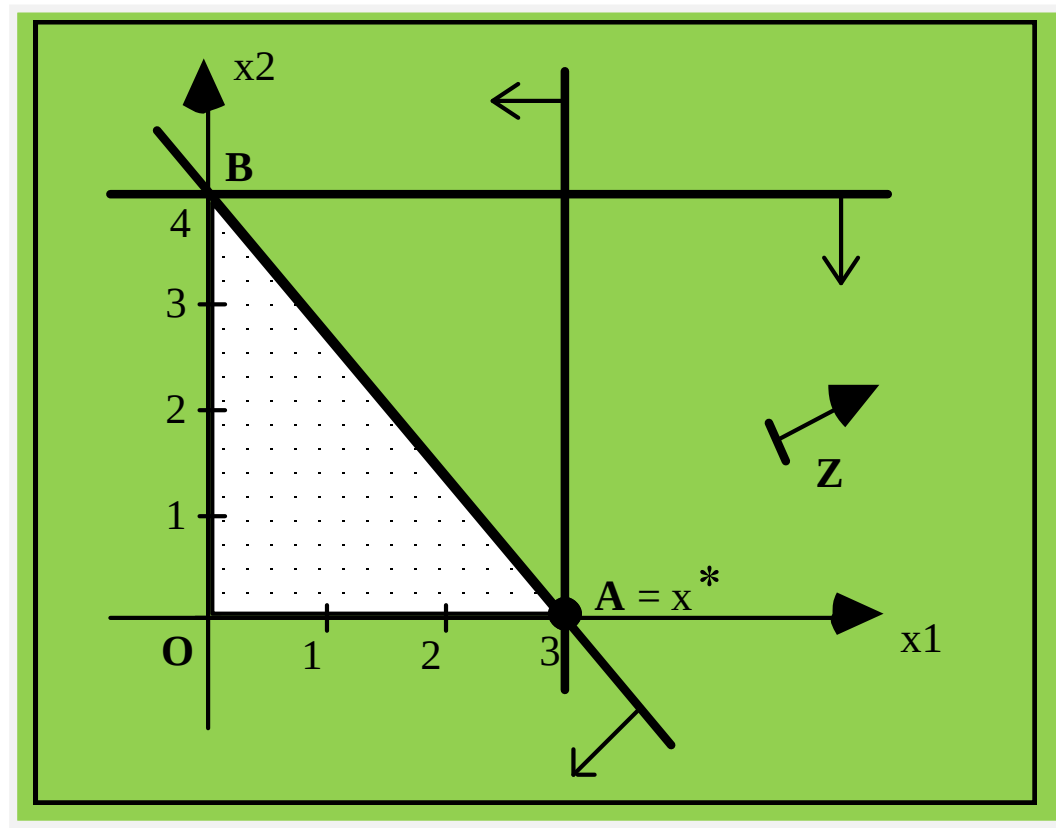
If the “Big M” technique was used, the same optimal table would be obtained (no negative values in the row $z_j - c_j$) with artificial variables in the basis and the conclusion to take would be just the same.

Sugestion: Try to solve the same problem using the “Big M” technique.

Degenerate solution

Example

$$\begin{aligned} \max \quad & z = 5x_1 + 2x_2 \\ \text{subject to} \quad & x_1 \leq 3 \\ & x_2 \leq 4 \\ & 4x_1 + 3x_2 \leq 12 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$



By the Simplex method:

$$\max z = 5 x_1 + 2 x_2$$

subject to

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 4$$

$$4 x_1 + 3 x_2 + x_5 = 12$$

$$x_i \geq 0; i = 1, 2, \dots, 5$$

Initial table

C_i		5	2	0	0	0	b	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5		
(?) x_3	0	1	0	1	0	0	3	$x_1 = 0$
x_4	0	0	1	0	1	0	4	$x_2 = 0$
(?) x_5	0	4	3	0	0	1	12	$x_3 = 3$
$Z_j - C_j$		-5	-2	0	0	0	0	$x_4 = 4$
		↑						$x_5 = 12$
								$Z = 0$

POINT O

We can remove x_3 or x_5 from the basis:

- If we remove x_3 then x_5 will become zero
- If we remove x_5 then x_3 will become zero

→ in both situations $x_3 = x_5 = 0$

Removing x_3 from the basis:

ci		5	2	0	0	0	b	
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	x_5		
x_1	5	1	0	1	0	0	3	$x_1 = 3$
x_4	0	0	1	0	1	0	4	$x_2 = 0$
← x_5	0	0	<u>3</u> *	-4	0	1	0	$x_3 = 0$
$Z_j - C_j$		0	-2	5	0	0	15	$x_4 = 4$
		↑						$x_5 = 0$
								$Z = 15$

POINT A

By the previous table we verify that the solution is not optimal yet $\rightarrow$ variable x_2 has to enter the basis.

However, by the graphic, we see that the solution is already the optimal one (POINT A)

Putting x_2 in the basis we get:

c_i		5	2	0	0	0		
$x_B \quad c'_B \quad x_i$		x_1	x_2	x_3	x_4	x_5	b	
x_1	5	1	0	1	0	0	3	$x_1 = 3$
x_4	0	0	0	$4/3$	1	$-1/3$	4	$x_2 = 0$
x_2	2	0	1	$-4/3$	0	$1/3$	0	$x_3 = 0$
$Z_j - c_j$		0	0	$7/3$	0	$2/3$	15	$x_4 = 4$
								$x_5 = 0$
								$Z = 15$

Optimal solution $\rightarrow$ POINT A

But this is the previous solution



Degenerate optimal solution

Removing x_5 from the basis we get:

c_i		5	2	0	0	0		
$x_B \quad c'_B \quad x_i$		x_1	x_2	x_3	x_4	x_5	b	
x_3	0	0	-3/4	1	0	-1/4	0	$x_1 = 3$
x_4	0	0	1	0	1	0	4	$x_2 = 0$
x_1	5	1	3/4	0	0	1/4	3	$x_3 = 0$
$Z_j - c_j$		0	7/4	0	0	5/4	15	$x_4 = 4$
								$x_5 = 0$
								$Z = 15$

Optimal solution → POINT A

Alternative optimal solutions

Example

$$\min z = x_1$$

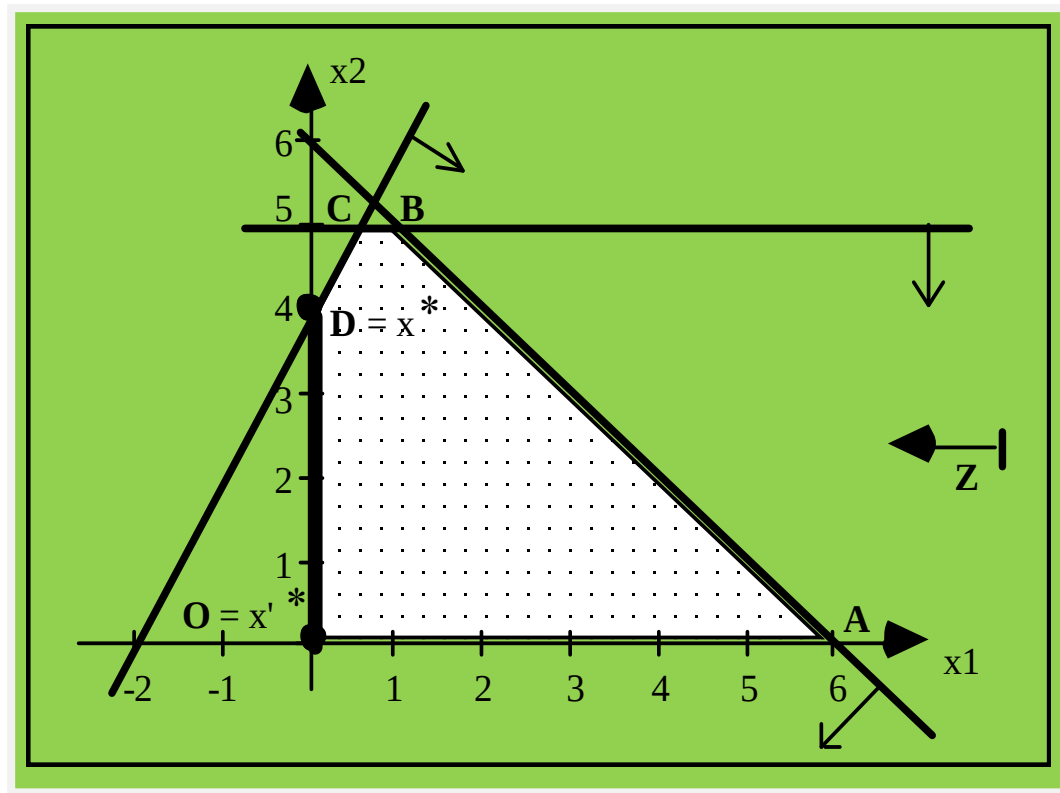
subject to

$$-2x_1 + x_2 \leq 4$$

$$x_1 + x_2 \leq 6$$

$$x_2 \leq 5$$

$$x_1 \geq 0, x_2 \geq 0$$



Solving by the Simplex method:

$$\max z = -x_1$$

subject to

$$-2x_1 + x_2 + x_3 = 4$$

$$x_1 + x_2 + x_4 = 6$$

$$x_2 + x_5 = 5$$

$$x_i \geq 0; i = 1, 2, \dots, 5$$

Initial table

C_i		-1	0	0	0	0	b	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5		
$\leftarrow x_3$	0	-2	<u>1</u> *	1	0	0	4	$x_1 = 0$
x_4	0	1	1	0	1	0	6	$x_2 = 0$
x_5	0	0	1	0	0	1	5	$x_3 = 4$
$Z_j - C_j$		1	<u>0</u>	0	0	0	0	$x_4 = 6$
			$\uparrow$					$x_5 = 5$
								$Z = 0$

Optimal solution $\rightarrow$ POINT O

But if variable x_2 goes to the basis there is no profit or prejudice!

➔ **alternative optimal solutions**

Putting x_2 in the basis, we reach the other alternative optimal extreme point:

C_i		-1	0	0	0	0	b	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5		
← x_2	0	-2	1	<u>1</u> *	0	0	4	$x_1 = 0$
x_4	0	3	0	-1	1	0	2	$x_2 = 4$
x_5	0	2	0	-1	0	1	1	$x_3 = 0$
$Z_j - C_j$		1	0	<u>0</u>	0	0	0	$x_4 = 2$
								$x_5 = 1$
								$Z = 0$

Optimal solution ➔ POINT D

The points **O** and **D** are optimal solutions for the problem, but also all the points that belong to the line segment linking them (which are convex linear combination of the referred two points).